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01410 Cryptology 1

Homework 3

Exercise 3.1

This exercise is about factoring integers. A square in $\mathbb{Z}$ is an integer x such that $x = y^2$ for another integer $y \in \mathbb{Z}$.

1. Suppose that $q - p = 2d > 0$, and $n = pq$, where d, p, q are integers. Show that $n + d^2$ is a square (in $\mathbb{Z}$).

Using the formulas for squaring a binomial, we obtain

$$n + d^2 = pq + \left(\frac{q-p}{2}\right)^2 = pq + \frac{q^2}{4} + \frac{p^2}{4} - \frac{pq}{2} = \frac{q^2}{4} + \frac{p^2}{4} + \frac{pq}{2} = \left(\frac{p+q}{2}\right)^2.$$

Now, $p + q = 2d + 2p$, so $(p + q)/2 \in \mathbb{Z}$, showing that $n + d^2$ is a square in $\mathbb{Z}$.

2. Given an integer n which is the product of two odd primes, and a given small integer d , defined as in 1. such that $n + d^2$ is a square (in $\mathbb{Z}$), show how this information can be used to factor n .

Since $d > 0$ is a small integer given as $(q - p)/2$, the primes p, q must lie close together (the difference $q - p = 2d$ is small) and $q > p$. Since

$$n = \left(\frac{q+p}{2}\right)^2 - d^2, \tag{1.1}$$

the integer $u = (p + q)/2$ can only be slightly larger than $\sqrt{n}$ and $u^2 - n$ is a square in $\mathbb{Z}$. This means that we can just try

$$u = \lceil \sqrt{n} \rceil + k, \quad k = 0, 1, 2, \dots$$

until u becomes a square in $\mathbb{Z}$. Then $q = u + d$ and $p = u - d$ (since $n = pq$ with $q > p$ and because of equation (1.1)).

3. Use the technique in 2. to factor $n = 2189284635403183$.

First, we find $\sqrt{n} \approx 46789791.99\dots$, so trying $k = 0$, we find $u = 46789792$ and

$$d = \sqrt{u^2 - n} = \sqrt{46789792^2 - 2189284635403183} = 9 = 3^2,$$

which is a square in $\mathbb{Z}$. Hence $q = 46789801$ and $p = 46789783$, which – due to their size – are easily checked to be primes.

Exercise 3.2

Let $n = 2051150372907037$ and define the hash function

$$H_F(m) \equiv 39^m \pmod{n}$$

for $m \in \mathbb{Z}$. Let $p = 2189284635404723$, which is a prime, and where $(p - 1)/2$ is also a prime.

1. Find a primitive element $\alpha \in (\mathbb{Z}/p)^*$ and choose a private key $a \in \mathbb{Z}/(p-1)$ such that $\text{DLP}(p)$ is difficult. Show your calculations.

Since $(p-1)/2$ is prime, the only prime factors in $|(\mathbb{Z}/p)^*| = p-1$ are 2 and $(p-1)/2$. Since the order of an element in $(\mathbb{Z}/p)^*$ must divide $p-1$ (Theorem 5.1.1 uses this fact), the only possible orders are 1 (the identity), 2 (the element -1), $(p-1)/2$ and $p-1$ (the primitive elements). Hence, any element $\alpha \not\equiv \pm 1 \pmod{p}$ such that $\alpha^{(p-1)/2} \not\equiv 1 \pmod{p}$ must have order $p-1$, i.e. be primitive (this is contained in Corollary 5.1.1). Taking $\alpha = 42$ and testing it in MAPLE¹ using the command

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p:=2189284635404723: 42^((p-1)/2) mod p;
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we see that $42^{(p-1)/2} \not\equiv 1 \pmod{p}$ and clearly $42^2 \not\equiv 1 \pmod{p}$. Hence 42 has order $p-1$, thus being primitive in $(\mathbb{Z}/p)^*$. $\text{DLP}(p)$ is independent of the choice of primitive element, $\alpha \in (\mathbb{Z}/p)^*$, because of the functional equation of the exponential function: Taking the primitive elements $\alpha_1, \alpha_2 \in (\mathbb{Z}/p)^*$ and a $\beta \in (\mathbb{Z}/p)^*$, we have

$$\alpha_1^{\log_{\alpha_1} \beta} = \beta = \alpha_2^{\log_{\alpha_2} \beta} = \left(\alpha_1^{\log_{\alpha_1} \alpha_2} \right)^{\log_{\alpha_2} \beta},$$

so $\log_{\alpha_2} \beta \equiv (\log_{\alpha_1} \beta)(\log_{\alpha_1} \alpha_2)^{-1} \pmod{p}$.

Since the only restrictions for $\text{DLP}(p)$ to be hard are on α and p , we choose the private key, $a \in \mathbb{Z}/(p-1)$ at random, here $a = 8928463543$.

2. Use α , a and p to set up the El-Gamal digital signature system. Let m be your 5-digit DTU student number (without the leading 0) regarded as an integer. Compute the signature of m using the El-Gamal system with the hash function H_F and the "random" number $k = 1234567$.

Once the hash function, H_F , the prime, p , the primitive element, α , the private key, a and the random number, k , we can set up the El-Gamal digital signature scheme.

The message is the 5-digit DTU student number, $m = 10269$. Hashing m with H_F , we obtain

$$\begin{aligned} H_F(10269) &\equiv 39^{10269} \\ &\equiv 494341953576409 \pmod{2051150372907037}, \end{aligned}$$

as our value of x . The signature $(\gamma, \delta) \in \mathbb{Z}/p \times \mathbb{Z}/(p-1)$ is then given as

$$\begin{aligned} \gamma &\equiv 42^{1234567} \\ &\equiv 2076571105570857 \pmod{2189284635404723} \\ \delta &\equiv (494341953576409 - 8928463543 \cdot 2076571105570857)(1234567)^{-1} \\ &\equiv -18540589410042258974189942 \cdot 427810349476471 \\ &\equiv 943318927577610 \pmod{2189284635404722}, \end{aligned}$$

i.e. $(\gamma, \delta) = (2076571105570857, 943318927577610)$. To find the multiplicative inverse of 1234567 in $(\mathbb{Z}/(p-1))^*$, one could also use Euclid's extended algorithm. We have used the command `1234567^(-1) mod 2189284635404722` in MAPLE.

¹All calculations are performed using MAPLE (the operator `&^` is used to force repeated squaring)

3. Show that the signature produced in the previous step will be verified as the signature on m .

To check that $(\gamma, \delta) = (2076571105570857, 943318927577610)$ is verified as the signature on m ,

$$(\alpha^a)^\gamma \gamma^\delta \equiv \alpha^x \pmod{p} \quad (1.2)$$

must hold.

Computing the factors of the left-hand side individually, we find

$$\begin{aligned} (\alpha^a)^\gamma &\equiv (42^{8928463543})^{2076571105570857} \equiv 1981556937118074 \pmod{2189284635404723} \\ \gamma^\delta &\equiv 2076571105570857^{943318927577610} \equiv 744839413869669 \pmod{2189284635404723}, \end{aligned}$$

$$\text{so } (\alpha^a)^\gamma \gamma^\delta \equiv 1968322644251032 \pmod{2189284635404723}$$

The right-hand side is

$$\alpha^x \equiv 42^{494341953576409} \equiv 1968322644251032 \pmod{2189284635404723}.$$

Thus, equation (1.2) is fulfilled, showing that $(\gamma, \delta) = (2076571105570857, 943318927577610)$ is verified as the signature on m .